

# Assignment – Part 2 Report

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## Introduction

The objective of this report is to document the development of a job application web of a mineral water company called DropFresh. The report will highlighting the key features, security measures, and personal contributions to the project.

The report will outline the main functionalities of the website, discuss the technical details of the implementation, and explore ways to improve the security of the web application. It will also reflect on the experience of working on this project and discuss relevant web development issues.

Youtube link: [https://youtu.be/m94GFYza\\_l0](https://youtu.be/m94GFYza_l0)

## Website Introduction

The website contains the following pages – Home Page, Career Opportunities, Job Application, About Us, Enhancement at Assignment 1, Enhancement at Assignment 2 and Manage.

### 1. Home Page

The home page introduces the company DropFresh and highlights its commitment to providing pure, refreshing mineral water. It outlines the key promises like refreshing taste, purity and sustainable practices. It also encourages visitors to learn more and join the brand community.



## Welcome to DropFresh - Pure Refreshment, Every Drop Counts!

### Discover Our Promise of Purity

#### Refreshing Taste, Pure Source:

DropFresh mineral water comes from pure springs located in untouched natural environments. The water goes through thorough filtration processes to ensure that it maintains its natural minerals and refreshing flavor.

#### Purity You Can Trust:

We recognize the significance of being transparent about the water you consume. That is why we offer full traceability for our water, enabling you to know its origin and the treatment process it undergoes at each stage.

#### Sustainable Development:

DropFresh is committed to protecting the environment by using eco-friendly packaging and sourcing products ethically. Our goal is to protect the planet for future generations while providing the cleanest water.

### Experience the DropFresh Difference



#### Become a member of the DropFresh family

Join "Stay hydrated" community

#### Follow us



At DropFresh, we're passionate about sustainability and protecting the environment. Learn more about our eco-friendly practices and our dedication to minimizing our environmental footprint for a healthier planet.

#### Contact Information

Email: [info@dropfresh.vn](mailto:info@dropfresh.vn)

Phone: 0849026110

Address: A35, Bach Dang, Tan Binh District, Ho Chi Minh City

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## 2. Career Opportunities

This page lists the open job positions at DropFresh. It includes a brief overview of each role and a link to view the detailed job description. The descriptions highlight the responsibilities, required skills, benefits etc.

[Enhancement Assignment 1](#) [Enhancement Assignment 2](#) [Hotline 0849026110](#) [Contact Us](#)

 [Home Page](#) [Career Opportunities](#) [Job Application](#) [About Us](#)



# Refresh your career at DropFresh

### Open Positions



#### DATA SCIENTIST

Location: Ho Chi Minh    Number of position: 1  
Posted: 16/02/2018    Deadline to apply: 31/03/2018

[View more](#)



#### IT SUPPORT SPECIALIST

Location: Ho Chi Minh    Number of position: 1  
Posted: 16/02/2018    Deadline to apply: 31/03/2018

[View more](#)



#### MARKETING MMANAGER

Location: Ho Chi Minh    Number of position: 1  
Posted: 16/02/2018    Deadline to apply: 31/03/2018

[View more](#)



#### SOFTWARE ENGINEER

Location: Ho Chi Minh    Number of position: 1  
Posted: 16/02/2018    Deadline to apply: 31/03/2018

[View more](#)

Join our team and be part of the DropFresh journey! We are a leading mineral water brand committed to providing pure, refreshing hydration to our customers. As we continue to grow, we are seeking talented individuals to join our dynamic team.



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This is where you can click in View more to see detail job description, also including a shortcut button leading to the Apply from

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# Refresh your career at DropFresh



**CLOSE**

**POSITION TITLE:** Data Scientist**POSITION ID:** DF01D

**JOB DESCRIPTION**

We are looking for an experienced Data Scientist to analyze data and develop models to extract meaningful insights for our business.

**JOB BENEFIT**

 Laptop

 Insurance

 Travel opportunities

 Allowances

 Uniform

 Incentive bonus

 Training & Development

 Salary review

 Health checkup

 Seniority Allowance

 Annual Leave

 Health checkup

**SALARY RANGE:** \$90,000 - \$120,000

**REPORTS TO:** Analytics Manager

**KEY RESPONSIBILITIES**

Mine data from databases and compile relevant information for analysis  
Develop and implement statistical models, machine learning algorithms, and other advanced techniques  
Visualize and present data findings to stakeholders  
Identify trends and patterns in data sets  
Make recommendations for process improvements based on analysis  
Maintain data systems and databases

**REQUIRED SKILLS AND QUALIFICATIONS**

Location: Hanoi city

Number of position: 1

Posted: 10/02/2024

Deadline to apply: 31/03/2024



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### 3. Job Application Form

The Apply page contains a form that allows candidates to submit their application for a specific job role. It collects all the necessary information like name, contact details, skills etc. The form content is structured using HTML5 input fields like text, email, radio buttons, dropdowns etc.

Form with interfrity checked, where you will receive error message if you fill in the information that doesn't match the requirement.

The data will be sent into the database, auto create new table in case the record table have not exist yet.

[Enhancement Assignment 1](#)[Enhancement Assignment 2](#)[Hotline 0849026110](#)[Contact Us](#)

[Home Page](#)[Career Opportunities](#)[Job Application](#)[About Us](#)

## Job Application Form

Please carefully check the information filled in before sending

\*Job Reference Number:  
(5 alphanumeric characters)

\*First Name:  
(max 20 alpha characters)

\*Last Name:  
(max 20 alpha characters)

\*Date of Birth:

\*Gender:

Please select -

\*Street Address:  
(max 40 characters)

\*Suburb/Town:  
(max 40 characters)

\*State:

Please select -

\*Postcode:  
(exactly 4 digits)

\*Email:

\*Phone Number:  
(8 to 12 digits)

\*Skills:

☐ HTML ☐ CSS ☐ JavaScript ☐ Other skills...

Other Skills:

Enter Here

APPLY NOW



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#### 4. About us

This page introduces me with my details like name, ID, skills etc. It also contains my timetable in a table structure. The layout uses HTML elements like definition lists, figures, and anchor links.

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[Home Page](#)[Career Opportunities](#)[Job Application](#)[About Us](#)

### About Our Group

**My Name:** Truong Dong Nhi

**My ID:** SWS00650

**Tutor's Name:** Mr. Le Duc Hieu

**Course:** COS10026-Computing Technology Inquiry Project

**My Programming Skill:** Python, ASM, Ruby

**Email to me:** 104977535@student.swin.edu.au



### My Timetable

| Time           | Monday          | Tuesday         | Wednesday       | Thursday        | Friday          |
|----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| 9:50am-11:30am | Web Design      | Web Design      | Web Design      | Web Design      | Web Design      |
| 2:00pm-11:30pm | Web Programming | Web Programming | Web Programming | Web Programming | Web Programming |



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## 5. Enhancement for Assignment part 1

Including 2 css enhancement with detail describe

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**This is the auto hover feature. This feature is intended to attract users to the information I want to highlight.**  
For this feature I use animation: autohover and autohover keyframes. Autohover keyframes are used to format the display for it when autohover. And in this button I have added a stop state when the mouse is inserted. Besides, I want users to be more certain about their choice, I have set the font size to be larger when hovering to clearly see the difference.

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**This is a feature when clicking 'view more', a new table will be displayed right on the interface without having to redirect to another page.**  
This helps users save time, they can directly view the jobs they want to apply for. In terms of code that implements this feature, in the default state, the website hides the content of the job description, this job description has an id in the general class. Then I use href="#id that I placed in 'view more'. The most important part is the css so that the hidden class displays:block when targeted. Besides, I also have a close button at the top, when I click on it, I want it to display the jobs.html page again and target the id of that job. If it's just a simple link of the a tag, it will jump to the top of the page and make the user experience not good because they has to scroll down again.



**DATA SCIENTIST**

[View more](#)

Location: Ho Chi Minh

Number of position: 1

Posted: 16/03/2014

Deadline to apply: 31/03/2024

  
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## 6. Enhancement for Assignment part 2

Including 2 php enhancement with detail describe

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### Enhancement 2

This feature serves as a confirmation mechanism prior to accessing the management page, ensuring that only authorized managers are granted access. When attempting to access the management page without being logged in, users will be redirected to the login page to authenticate their identity and role. Once logged in, they are then allowed access to the management page [Click here to jump to the function](#)

## Login

Username:  Password:  [Login](#)

After accessing the management page and performing necessary tasks, users can choose to logout. Upon logout, access to the management page will be restricted. Any attempt to access the management page directly via a link without proper authentication will result in redirection to the login page. This ensures that only authenticated users with valid credentials can access the management functionality [Click here to jump to the function](#)

[Logout](#)

In addition to authentication and logout features, the management system includes sorting and searching capabilities. Users can sort the list of IDs in ascending or descending order. Furthermore, they can search for entries based on first name, last name, or status. Additionally, users have the option to perform alphabetical sorting, facilitating easier navigation and retrieval of data within the system. [Click here to jump to the function](#)

Sort by:  Sort order:

[Logout](#)



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## 7. Manage

This site can't be access directly via URL, it only can be reach when you login (*Username: Management, Password: 104977535*)

The EOI will have various way to interaction, according to the requirement as:  
List all EOIs.

- List all EOIs for a particular position (given a job reference number).
- List all EOIs for a particular applicant given their first name, last name or both.
- Delete all EOIs with a specified job reference number
- Change the Status of an EOI.
- Sort the data in various field
- Logout button

Enhancement Assigned 1

Enhancement Assigned 2

Hotline: 0849028110

EN Control UI



[Home Page](#)[Career Opportunities](#)[Job Application](#)[About Us](#)

### View All Jobs Records

Reset All

Job ID to update status:

Job Reference number to delete:

Job Reference:  Job Reference Code:  First Name:  Last Name:

Sort By:  Job ID:  Sort order:

| Job ID | Job Reference Code | First Name | Last Name | Age | Age group | Gender | Suburb   | State | Postcode | Email                | Phone      | Address   | Status  | Action               |
|--------|--------------------|------------|-----------|-----|-----------|--------|----------|-------|----------|----------------------|------------|-----------|---------|----------------------|
| 1      | 01010              | Truong     | Dong Min  | 18  | -         | male   | Gach Hai | VIC   | 300      | intedhuc22@gmail.com | 0849028110 | Bach Dang | Current | <a href="#">View</a> |



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## Security of the website

- When inserting information, use the `bind_param` function to ensure that malicious SQL injection code is not inserted. This feature established in both apply form and manage page (wherever the user can fill input)

```
$stmt->bind_param("ssssssssss", $jobref, $firstname, $lastname, $date, $gender, $address, $birthdate, $state, $zipcode, $email, $phone, $hashedPassword, $skillsString, $otherSkills);  
$stmt->bind_param("ss", $username, $password);  
$stmt->execute();  
$stmtUpdateStatus->bind_param("si", $status_to_update, $update_eoi_number);  
$stmtDeleteJob = $conn->prepare($queryDeleteJob);  
$stmtDeleteJob->bind_param("s", $delete_jobref);  
$stmt->bind_param("s", $filterJobrefValue);  
$stmt->execute();  
$input = stripslashes($input); (this is similar with bind-param)
```

- Sanitize the data: the process of removing or replacing unwanted or dangerous characters from user data to prevent attacks or errors in web applications

```
$jobref = sanitize($_POST['jobref']);
```

- Prevents Cross-Site Scripting (XSS) attacks by causing special characters not to execute as HTML code but instead to be displayed as regular text

```
<p class="container-p"><strong>Job Reference Number:</strong> ' . htmlspecialchars($jobref) . '</p>
```

## My contribution

I researched, designed and built all of the above features by myself.

## Reflection and Discussion

Through this project, I have gained valuable experience in developing a server-side web application that integrates a database for storing and managing user-submitted data. It has reinforced my understanding of the importance of modular and reusable code, as well as the need for thorough data validation and security measures to protect the integrity of the system.

One of the key challenges I faced was ensuring that the server-side validation was robust and effective, even in cases where the client-side validation could be bypassed. I spent a significant amount of time researching common validation techniques and implementing checks to handle various edge cases. This experience has strengthened my appreciation for the role of server-side validation in maintaining data quality and security.

Another aspect of the project that I found interesting was the consideration of privacy and security issues. To address this, I implemented the login system for the HR manager and ensured that the database connection details were stored securely. Moving forward, I will be more mindful of these types of security and privacy concerns when designing and developing web-based systems.

The experience of working on this project as an individual has also helped me improve my time management, problem-solving, and technical documentation skills. I had to carefully plan and prioritize my tasks, address any challenges that arose, and thoroughly document my work.

Overall, this project has been a valuable learning experience that has deepened my understanding of server-side web development and the importance of considering non-technical aspects when building web applications. I am confident that the skills and knowledge I have gained will be highly applicable in my future endeavors.

## Conclusion

This project has provided me with valuable experience in server-side web development and database integration. I have gained a deeper understanding of the importance of modular and reusable code, as well as the critical role of thorough data validation and security measures. The challenges I faced in implementing robust server-side validation have strengthened my appreciation for holistic approaches to data integrity. Additionally, this project has helped me develop a more nuanced understanding of the non-technical considerations, such as privacy and security, that must be addressed when building web applications. The experience of working on this project individually has also enhanced my time management, problem-solving, and technical documentation skills. Overall, this project has been a transformative learning experience that has significantly expanded my knowledge and capabilities in server-side web development, which I will apply in my future endeavors.

## Reference

- Official PHP Documentation: The official PHP website - [php.net](https://www.php.net) - offers a lot of documentation and tutorials on the language, including related topics like form handling, database queries, and more.

- Tutorials Point: The website [tutorialspoint.com/php](https://www.tutorialspoint.com/php) has many detailed PHP tutorials, including source code examples.
- W3Schools: The website [w3schools.com/php](https://www.w3schools.com/php) offers basic to advanced PHP lessons, with illustrative examples.
- Sitepoint: The website [sitepoint.com/php](https://www.sitepoint.com/php) has many articles, tutorials and tutorials on PHP programming.
- Codecademy: The website [codecademy.com](https://www.codecademy.com) offers an interactive course on PHP programming.
- Udemy: The website [udemy.com](https://www.udemy.com) has many courses and tutorials on PHP, from basic to advanced.
- Stackoverflow: The website [stackoverflow.com](https://stackoverflow.com) is a great resource to find answers to PHP-related problems.